

NFC-enabled credit or debit card and a payment terminal. The data is then used to create a copy of the card. Or hackers can use the high-range NFC reader and writer and pass by you to read the card details and make a copy of your NFC card to make fraudulent purchases.

There are various tools and apps like Flipper using which the hackers can read your card data, store it, and even clone it. The data stored in NFC can also be copied easily by a phone or a reader device. The hackers can then use your card number and other details to use it for fraudulent purchases. In worst cases, hackers can even inject their own code or simply erase your credit card by using NFC writer tools. However, no such complaints have arisen yet.

It is however hard nowadays to do fraudulent transactions using a hacked credit or debit card details as it requires an OTP or PIN while spending beyond a certain amount (generally, ₹5,000). A couple of ways a hacker may copy your card's details are mentioned below.

Man-in-the-middle attack. This involves a hacker intercepting the communication between an NFC-enabled card and a payment terminal, and either altering the data being transmitted or injecting their own data into the communication.



Faraday Bag for credit and NFC cards

Physical tampering. This involves physically modifying an NFC-enabled card or payment terminal in order to bypass security measures or to gain access to sensitive data.

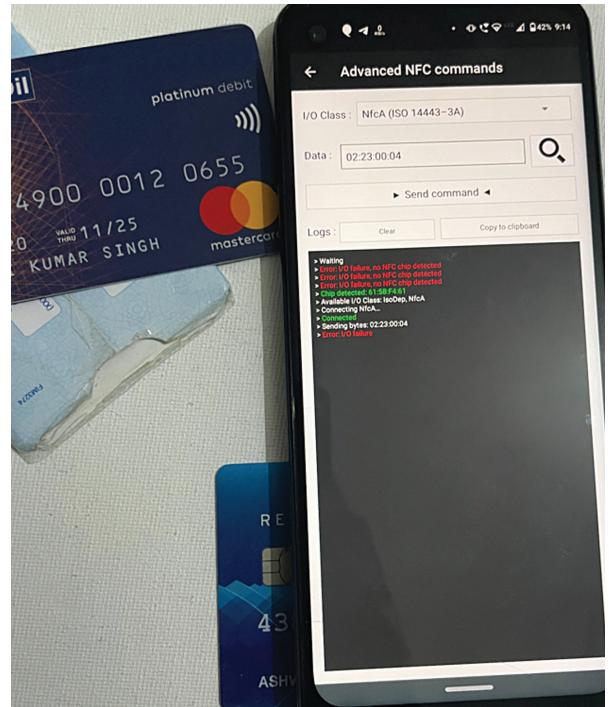
The main feature of NFC is it works only over a very short distance. So, hackers can fetch data from your NFC-enabled card only when they are near the card. It is possible to design an NFC reader with a longer range by using multiple coils.

NFC operates at a high frequency 13.56MHz and requires a close proximity between the two devices in order to establish a connection. Increasing the range of an NFC reader beyond a few centimetres would require the use of a lower frequency, which would result in a slower data transfer rate.

If you analyse the underlying physical equations in the case of a circular loop coil antenna, the near field range (suitable for NFC communications) is about 1.414 times the radius of the circular coil. To achieve a practical 60cm range, you may try to create a near field with a theoretical range several times bigger, for instance, 180-240cm, for which you need a circular coil of about 2.5-3.4 metres in diameter. But you can change the shape and try the square shape for that.

How to protect yourself

The possibility of your debit or credit card being cloned increases tremendously in public places and crowded metro, bus,



The author tests the injection of codes and attempt to change the information inside the NFC of the credit card

train, etc, while travelling. Anyone with an NFC reader or phone can get your card data by coming close to your wallet in your pocket or elsewhere.

One of the measures that can be taken to protect yourself is by using the Faraday purse or wallet. These wallets have layers of Faraday cage that block the signal and protect against your cards' data being read or cloned in such crowded places. There are some small metallic enclosures also available, which can be easily kept in a pocket like a wallet, to keep the cards protected.

Setting limit to NFC transactions can also protect you from heavy losses. You can ask your bank or card provider to set transaction limit for contactless payments through your cards. Using, multi-factor authentication or a digital PIN setting for an NFC card is another way to protect yourself from fraud. **EFY**

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